



Working with the o9 platform

ABSTRACT

Log In to a tenant, create modelling constructs, and execute IBPL rules on the powerful o9 Platform. Enhance your model with widgets available in the platform.

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Tenants

Logging In

- 1) Open Google Chrome web browser
- 2) Type the URL <https://mytraining.o9solutions.com>
- 3) For first time login:
 - a. Type in your User ID ([firstname.lastname@company.com](#))
 - b. Type in the SSO password
 - c. Click **Sign In/Login**.
- 4) Check the **Stay signed in** check box.
- 5) Click **Sign in**.
- 6) Click the **Workspaces**  button at the top left to open the workspaces navigation pane.
- 7) Open the **Tenants** workspace.
- 8) Select the **Manage Tenants** tab
- 9) Click the arrow to the left of the tenant information row to see the list of users.
- 10) Select the **Users** tab to list the users for this tenant.
- 11) Your User ID should already be added to the list.

You have successfully logged in as administrator to your newly created tenant.

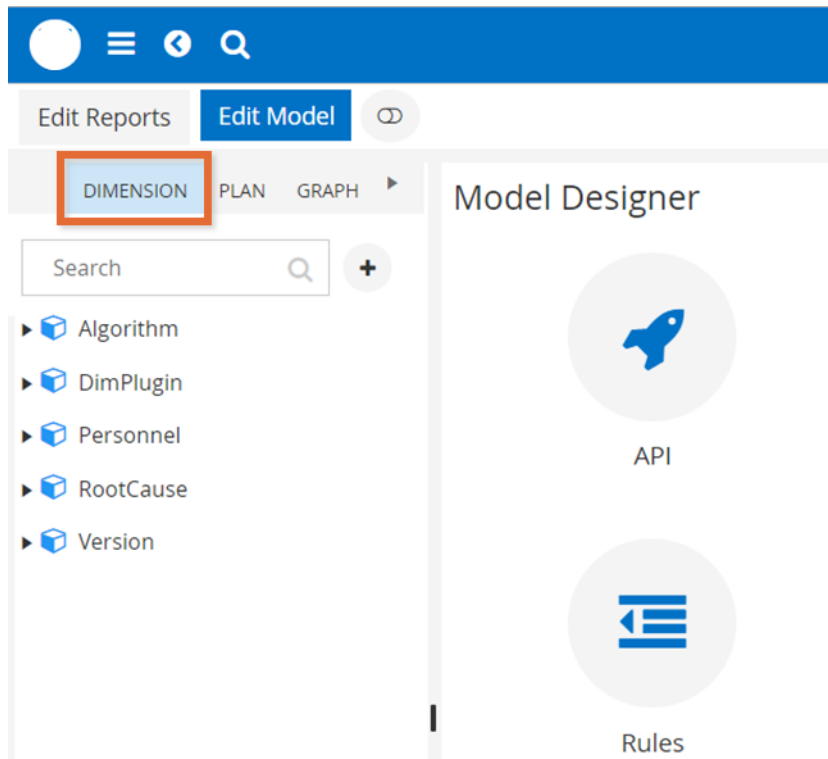
Dimensions

Definition of dimension: A Dimension is a name given to identify a set or group of data that usually represents Master data. It can be compared to a table name.

Adding Dimensions

You will be creating Four dimensions: Product, Customer, Region, and Time.

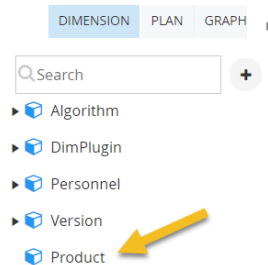
- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable Model Designer.
- 3) Make sure the **Dimension** tab is selected under the **Edit Model** button.



- 4) Click the button to the right of the Search box to create a new Dimension object.
- 5) Populate the form fields:

Name	Product
Description	A product
Time Like Dimension	FALSE (leave unchecked)

- 6) Click the button at the top of the form.
- You should now have a Dimension called Product.



Repeat the process to create Customer, Region, and Time dimensions using these values:

Name	Customer
Description	Customer data

Time Like Dimension	FALSE (leave unchecked)
----------------------------	-------------------------

Name	Region
Description	Region data
Time Like Dimension	FALSE (leave unchecked)

Name	Time
Description	Time
Time Like Dimension	TRUE (check the box). You MUST check this box, as you cannot change this setting later

There should be four dimensions added – Product, Customer, Region, and Time.

Level Attributes

Level attributes are attributes of the dimension that are used to group the dimension data for aggregation or serve as their unique key codes. You will now configure level attributes for each of the dimensions you have created. You will start with the **Product** dimension.

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window.
- 3) Select the **Dimension** tab under the **Edit Model** button.
- 4) Select **Product** dimension.
- 5) Click the  button in the **Attributes** pane to add a new attribute.
- 6) Create the SKU attribute, which is the unique key for the Product. Enter the following values in the form:

Name	SKU
Description	A unique name of this product
Order By	Name
Leaf	TRUE (Check the box for key attribute)
Seed Tags	TRUE (Ensure the box is checked)
Key Column Data Type	Integer
Key Generation	Auto
Show Filter As...	Tile View

- 7) Once the form is complete, click the button at the top of the form. You should now see a new SKU attribute in the Attributes pane.
- 8) Continue this process to create these attributes for the associated dimensions:

Product (Total of 4 attributes)

Customer (Total of 3 attributes)

Month

Name *

Month

Description *

Month

Order By

Key

☒ Leaf

☒ Seed Tags

Key Column Data Type

Datetime

Key Generation

UserSupplied

Show Filter As...

Tile View

Aliases

No content. Add content using add button

Quarter

Name *

Quarter

Description *

Quarter

Order By

Key

☐ Leaf

☒ Seed Tags

Key Column Data Type

Datetime

Key Generation

UserSupplied

Show Filter As...

Tile View

Aliases

No content. Add content using add button

Year

Name *

Year

Description *

Year

Order By

Key

☐ Leaf

☒ Seed Tags

Key Column Data Type

Datetime

Key Generation

UserSupplied

Show Filter As...

Tile View

Aliases

No content. Add content using add button

YTD

Name *

YTD

Description *

Year To Date

Order By

Name

Leaf

Seed Tags

Key Column Data Type

Datetime

Key Generation

UserSupplied

Show Filter As...

Tile View

Aliases

No content. Add content using add button

The dimensions Product, Customer, and Time should have level attributes as shown below:

DIMENSION
PLAN
GRAPH

Search

Algorithm

Customer

Account

All Accounts

Channel

DimPlugin

Personnel

Product

All Products

Classification

Product Category

SKU

Time

Month

Quarter

YTD

Year

Version

You will create the level Attributes for the Region dimension later in the exercise.

Attribute Properties

Attribute properties are properties that are used for filtering or reporting but not for grouping and aggregation. You will add properties to some of the level attributes you have created.

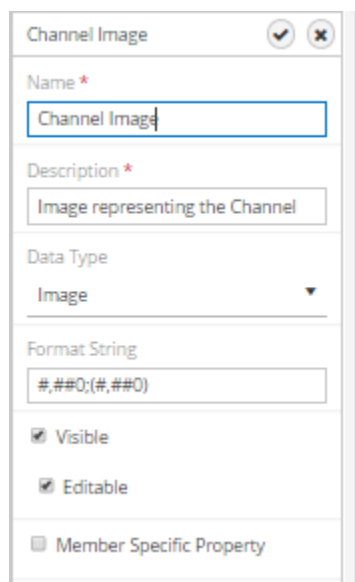
- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window.
- 3) Select the **Dimension** tab under the **Edit Model** button to enable the Model Designer.
- 4) Click the arrow next to the Customer dimension to expand the list of level attributes underneath it.
- 5) Click the Account level attribute.
- 6) You should see four panes, including a pane titled **Attribute Properties**. Click the  sign at the top of the Attribute Properties pane.
- 7) Enter the following values:

Name	Account Image
Description	Image representing the account
Data Type	Image

- 8) Once the form is complete, click the  button at the top of the form.

Continue this process to create the rest of the attribute properties using these values:

Customer.Channel



Channel Image

Name *

Channel Image

Description *

Image representing the Channel

Data Type

Image

Format String

###0;(##,##0)

☒ Visible

☒ Editable

☐ Member Specific Property

Product.Classification

Edit Reports

Edit Model

Adding property

Name *

Classification Image

Description *

An image representing this Classific

Data Type

Image

Format String

###0;(###0)

☒ Visible

☒ Editable

☐ Member Specific Property

Product.Product Category

Edit Reports

Edit Model

Adding property

Name *

Product Category Image

Description *

Image representing product categor

Data Type

Image

Format String

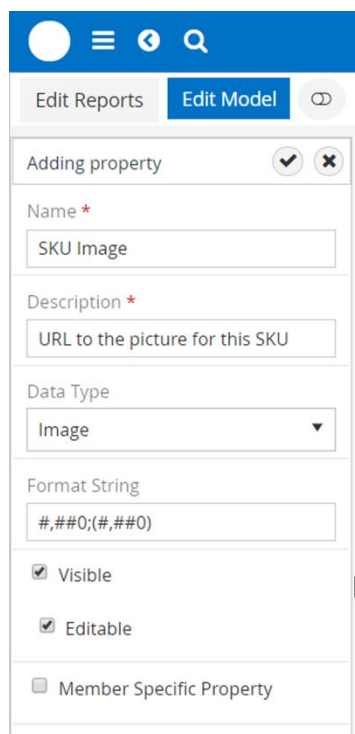
###0;(###0)

☒ Visible

☒ Editable

☐ Member Specific Property

Product.SKU



Hierarchies

A hierarchy is an ordered collection of levels, where the members of one level are parents to members of the next level. For example, an individual SKU can belong to a Product Category, meaning every member of the SKU level has a parent member that is in the Category level. You will now configure the hierarchies in the model.

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable the Model Designer.
- 3) Select the **Dimension** tab under the **Edit Model** button.
- 4) Click the **Customer** dimension.
- 5) Create a new Hierarchy by pressing the  button at the top of the Hierarchies pane.
- 6) Enter the following values:

Name	Customer
Description	Customer hierarchy

- 7) Once the form is complete, click the  button at the top of the form.

Adding Hierarchy

Name *

Customer

Description *

Customer hierarchy

- 8) Drag Attributes one at a time from the Attributes pane to the new hierarchy to define the levels of the hierarchy. The attributes should appear in this sequence in the hierarchy:
 - a. All Accounts
 - b. Channel
 - c. Account
- 9) When you are done, the hierarchy should look like this:

Attributes

+

Account

All Accounts

Channel

Hierarchies

Customer

All Accounts

Channel

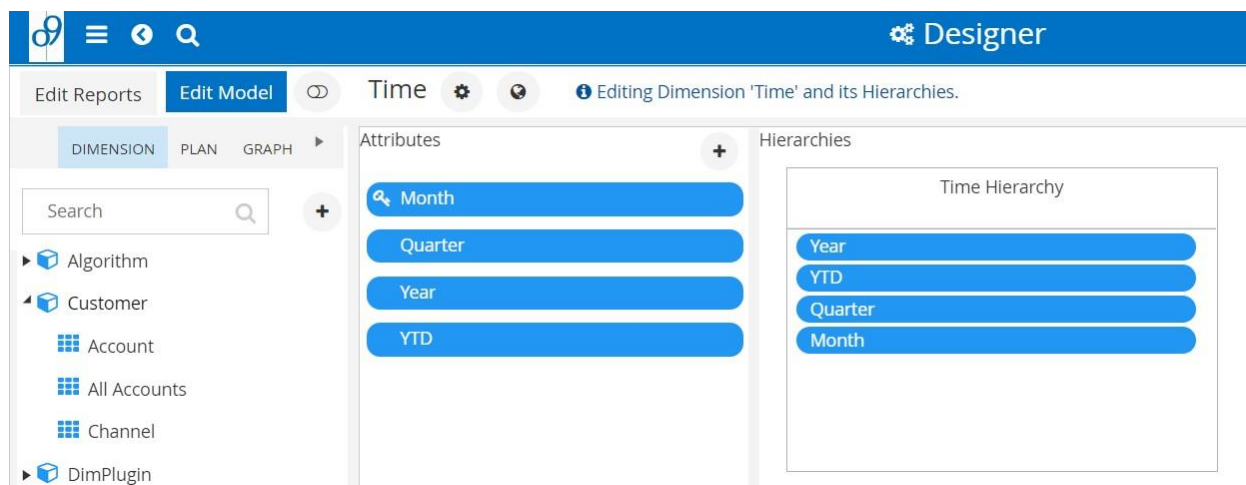
Account

- 10) When you are finished creating the Customer hierarchy, continue with these hierarchies:

Product

All Products
Classification
Product Category
SKU

Time



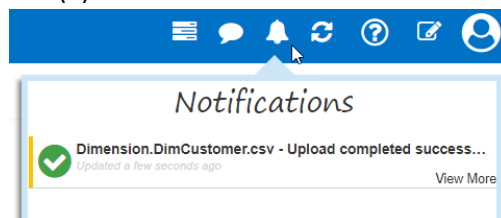
Importing Dimension Member Data

You will now import dimension member data using the Excel files that were provided with this document.



NOTE: Before uploading the member data, compare and confirm that both the Level Attribute names and the column headings in the provided Excel file are the same (case and space sensitive) for each Dimension.

- 1) Open the **Data Management** workspace.
- 2) Select the **Imports/Exports** view.
- 3) Click **Select Files**.
- 4) Locate the file *Dimension.DimCustomer.csv* that was provided for the class and click **Open**.
- 5) Click **Upload**.
- 6) Once the upload completes, the Notification  icon at the top right of the window will show a number indicating a new notification. Click the button and you should see confirmation that the customer data was loaded. Six (6) customer records were loaded.



- 7) Repeat these steps with the files *Dimension.DimProduct.csv* and *Dimension.DimTime.csv*.

You have now uploaded Customer, Product and Time data to the model. Next you will confirm the data was uploaded correctly.

- 8) Open the **Debugging** workspace.
- 9) Select the **Execute Query** view.

10) Paste this text into the Query form to view the customer data:

select ([Customer].[Account][Customer].[Channel]);*

11) Click **Execute**.

12) The results should look like this:

Account	Channel
Amazon	Mass Market
Future Group	Wholesale
Kroger	Mass Market
Metro	Wholesale
SAMS	Wholesale
Walmart	Mass Market

13) Paste this text into the Query form to view the product data:

select ([Product].[SKU][Product].[Classification]*[Product].[Product Category]);*

14) Click **Execute**.

15) The first few records of the query should return these results:

SKU	Classification	Product Category
Butterfinger	B	Chocolate
Cerelac	C	Baby Foods
Crunch	A	Chocolate
Gerber	A	Baby Foods
Gerber Graduates	C	Baby Foods
KitKat	B	Chocolate
Milo	C	Refreshment
Nestea	B	Refreshment

16) Finally, paste this text into the Query form to view the time data:

select ([Time].[Month][Time].[YTD]*[Time].[Year]);*

17) Click **Execute**.

18) The query should return 36 results, the first few of which should look like this:

Month	YTD	Year
01-01-2018	YTD-18	2018
02-01-2018	YTD-18	2018
03-01-2018	YTD-18	2018
04-01-2018	YTD-18	2018
05-01-2018	YTD-18	2018
06-01-2018	YTD-18	2018
07-01-2018	ROY-18	2018
08-01-2018	ROY-18	2018

Measures

Plans

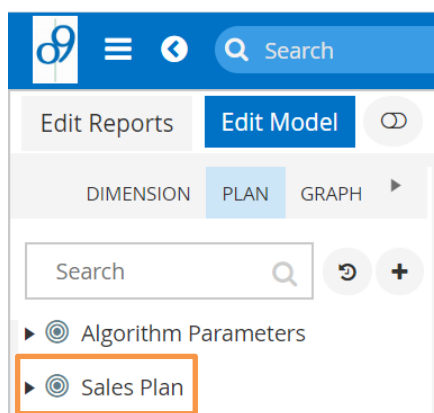
A plan consists of all the measure groups belonging to a business process.

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable Model Designer.
- 3) Select the Plan Tab.
- 4) Click the + icon to the right of the Search box to create a new Plan object.
- 5) Populate the form fields:

Plan Name	Sales Plan
Description	Measures for sales plan

- 6) Click the check mark in the circle at the top of the form.

You should now have a Plan labeled **Sales Plan**.



Measure Groups

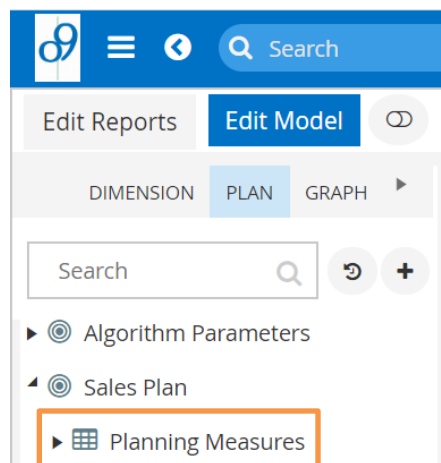
A measure group is a collection of measures that have the same granularity.

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable Model Designer.
- 3) Make sure the **Plan** tab is selected.
- 4) Click the  button to the right of the newly created Plan, called Sales Plan, to create a new measure group.
- 5) Populate the form fields:

Name	Planning Measures
Description	Group of measures used for sales planning

- 6) Click the  icon at the top of the form.

You should now have a Measure Group called **Planning Measures**.

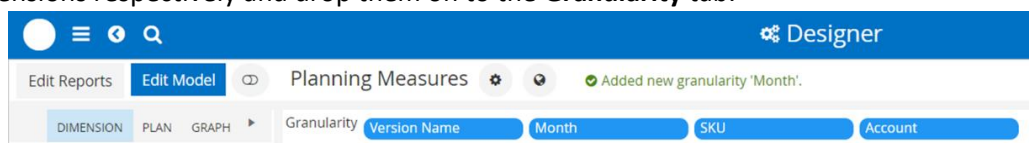


Measures

A measure is a measurement (also known as a metric or a fact) of a business process representing data that exists at the intersection of one or more dimensions. These are basically transactional data that will keep changing with time.

We will be creating five measures: Budget, Channel Inv, Consumption, Forecast, and Shipment

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable Model Designer.
- 3) Select the **Plan** tab.
- 4) Select the newly created measure group called Planning Measures and you would be able to see the **Granularity** tab with Version Name only.
- 5) Select the **Dimension** tab under the **Edit Model** Button.
- 6) Individually drag SKU, Account and Month level attributes from Product, Customer and Time Dimensions respectively and drop them on to the **Granularity** tab.



- 7) Select the **Plan** tab under the **Edit Model** button
- 8) Click the **+** button at the top of the Measures pane to create a new measure.
- 9) Populate the form fields:

Name	Budget
Description	Annual Operating Plan
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Sum

- 10) Click the **✓** button in the circle at the top of the form.

You should now have a measure called **Budget**.

Repeat the process for the Channel Inv, Consumption, Forecast, and Shipment measures using these values:

Channel Inv

Consumption

Name	Channel Inv
Description	Inventory at customer DC
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Sum

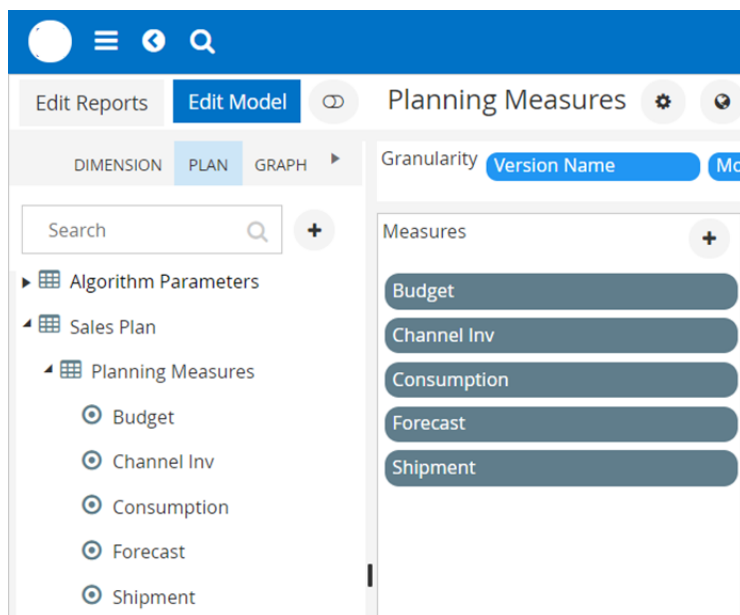
Forecast

Name	Consumption
Description	Units consumed at the customer DC
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Sum

Shipment

Name	Forecast
Description	Forecast of shipments
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Sum

Name	Shipment
Description	Units shipped from DC/Factory
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Sum



Importing data

- 1) Open **Data Management** workspace.
- 2) Click the **Imports/Exports** tab.
- 3) Click **Select Files** button and browse to the folder which has the measure data file (provided with this guide), in this case called *Fact.FactPlanningMeasures.(0)* on the local drive.
- 4) Click the **Upload** button.
- 5) A notification will pop up on the top right corner indicating whether the upload was successful or not
- 6) Open the **Debugging** workspace.
- 7) Select the **Execute Query** view.
- 8) Paste this text into the Query form to view the data:

```
select ([Version].[Version Name] * [Product].[SKU] * [Customer].[Account] * [Time].[Month] *
{Measure.[Budget], Measure.[Shipment], Measure.[Consumption], Measure.[Forecast]});
```
- 9) Click **Execute**.

The query should return 7560 results, the first few of which should look like this:

Version Name	SKU	Account	Month	Budget	Shipment	Consumption	Forecast
CurrentWorkin...	Butterfinger	SAMS	01-01-2018	7,555	8,363	7,910	
CurrentWorkin...	Cerelac	SAMS	01-01-2018	14,901	2,802	2,790	
CurrentWorkin...	Crunch	SAMS	01-01-2018	7,757	4,467	3,902	
CurrentWorkin...	Gerber	SAMS	01-01-2018	4,305	15,299	7,432	
CurrentWorkin...	Gerber Gradua...	SAMS	01-01-2018	14,640	10,008	10,073	
CurrentWorkin...	KitKat	SAMS	01-01-2018	1,346	12,475	13,264	
CurrentWorkin...	Milo	SAMS	01-01-2018	11,227	15,109	14,986	
CurrentWorkin...	Nestea	SAMS	01-01-2018	3,045	230	308	



NOTE: The measure data file must be uploaded ONLY after all the dimension data files have been uploaded.

Now that the data is uploaded into the Dimensions and the Measures, let us look executing some queries to view data using IBPL commands.

IBPL Commands

Using select command

To view a single Attribute Member –

```
select ([Time].[Year]);
```

```
select [Product].[SKU];
```

To view multiple Attribute Members –

```
select ([Product].[Product Category] * [Product].[SKU]);
```

```
select ([Time].[Quarter] * [Time].[Month]);
```

To select Measure data –

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Shipment], Measure.[Consumption]} ) limit 5;
```

```
select ([Version].[Version Name] * [Time].[Year].[2018] * [Time].[Month] * [Product].[SKU] *
{Measure.[Shipment], Measure.[Consumption]}); // The Shipment and Consumption of all SKUs for the
months belonging to the Year 2018
```

Filter Members using relatedmembers function –

```
select ([Product].[Product Category].filter(#.Name contains "Chocolate").relatedmembers ([SKU])); - //
Select all SKUs of a particular Category, such as Chocolate by using the relatedmembers function
```

```
select ( [Version].[Version Name] * [Product].[Product Category].[Chocolate].relatedmembers([SKU]) *
[Time].[Year].[2018].relatedmembers([Month]) * {Measure.[Shipment], Measure.[Budget]}); //Select all
```

the SKUs of the Category Chocolate, for all Months of the Year 2018 along with their Shipment and Budget values.

```
select ( [Version].[Version Name] * [Product].[Product Category].[Chocolate].relatedmembers([SKU]) *
[Time].[Year].filter(#.Name in {[2019],[2020]}).relatedmembers([Month]) * {Measure.[Shipment],
Measure.[Consumption]}); // Using the filter option select the SKUs of the Category Chocolate, for the
Months of Year 2019 and 2020 along with the Measure values
```

```
select ([Version].[Version Name] * [Time].[Year].[2018] * [Time].[Month] * [Product].[SKU] *
{Measure.[Shipment], Measure.[Consumption]}); // The Measure data of all SKUs for the months
belonging to the Year 2018
```

Filter using where

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Shipment], Measure.[Consumption]} )
where {[Product].[Product Category].[Refreshment]}
limit 5;
```

One can use multiple conditions such as member and measures together in the where clause. The final condition is a logical AND of all the conditions provided.

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Shipment], Measure.[Consumption], Measure.[Budget]} )
where {[Product].[Product Category].[Refreshment], Measure.[Shipment] > 7000,
Measure.[Consumption] > 8000};
```

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Shipment], Measure.[Budget]} )
where {[Time].[Year].filter(#.Name in {[2018], [2019]})};
```

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Consumption], Measure.[Shipment], Measure.[Budget]} )
where {[Product].[Product Category].[Chocolate]};
```

order using orderby

Measure data can be ordered by one measure either in ascending or descending order.
Example with descending order for one measure.

```
select ( [Version].[Version Name] * [Product].[SKU] * [Time].[Month] *
{Measure.[Shipment], Measure.[Consumption]} )
orderby Measure.[Consumption] desc;
```

One can also order by member properties. Note that in case of attributes, one needs to specify the respective order-by property selected during attribute definition (name or key or any other property) and not just level attribute.

```
select ([Version].[Version Name] * [Time].[Month] * [Product].[SKU] *
{Measure.[Shipment], Measure.[Consumption]})
orderby [Version].[Version Name].Name desc, [Product].[SKU].Name asc;
```

```
select ( [Version].[Version Name] * [Time].[Month] * {Measure.[Shipment] })
orderby [Time].[Month].Key;
```

Playing with the Time dimension level attributes

```
select Time.[Month].Filter(#. Name contains "4/1");
```

```
select Time.[Month].Filter(#. DisplayName contains "Apr");
```

```
select Customer.[Account].Filter(#. DisplayName contains "Kroger" );
```

```
select Customer.[Account].Filter(#.Key contains "2" );
```

```
select Customer.[Account].Filter(#.Key > "1" );
```

```
select Customer.[Account].Filter(#.Key == "3" );
```

```
select ([Customer].[Account].count * [Product].[SKU].count * [Time].[Month].count * [Version].[Version
Name].count );
```

```
Select Time.[Month].filter(#.Key >= todatetime("2019/07/01"));
```

```
select Time.[Month].filter(#.[Month$IsCurrent] == True);
```

Exercise

Write IBPL statements to view the below data

- 1) View the Budget, Channel Inventory, Consumption, Forecast, and Shipment for Account Walmart and Product Crunch
- 2) View the Product SKU and Measure details for the Month of April using the filter option
- 3) View the Measure details of the Account Kroger
- 4) View the Measure details whose Account has the Key as 2

- 5) Select all Accounts whose key is greater than 3
- 6) Select the Consumption, Budget and Shipment for all Products, month wise
- 7) Select the Budget and Shipment for the Account Walmart and Product Nestea
- 8) Display the Budget Value on a Yearly Basis
- 9) What are the total actuals for 2018?
- 10) Display the Products whose Shipment is greater than 10000
- 11) Select the Shipment for the months belonging to the Year 2018, for the category Baby Foods
- 12) What is the consumption of Crunch in SAMS for the year 2018?
- 13) How would you find out the SKUs which are part of Product Category 'Baby Food'?
- 14) What is the forecast for Chocolates for the month of Jul 2020?
- 15) Make a list of intersections where Channel Inventory is greater than 7000
- 16) What is the Channel Inventory for KitKat in Walmart?
- 17) What is the Yearly Consumption of C-class products in SAMS for 2018, 2019 and 2020?

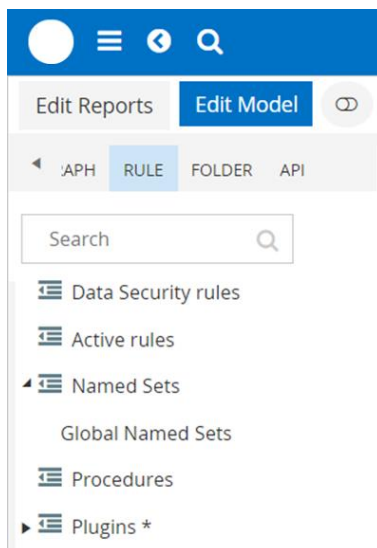
Named Sets

A named set is like a variable whose type is a Member Set. They can be alias names for a group or set of members. A named set is global and persists as long as the GraphCube server is running. Named sets are dynamic in the sense that the actual members in the list can change if members are created or deleted, or if attributes of the members change.

- 1) Open the **Designer** workspace.
- 2) Click the **Edit Model** button at the top left of the window to enable Model Designer.
- 3) Make sure the **Rule** tab is selected.
- 4) Click the  icon to the right of the Named Sets to add a new rule file.
- 5) Populate the form fields:

Label Name	Global Named Sets
Label Description	Named sets used globally in the tenant

- 6) Click the  button at the top of the form. You should now have a new file under Named Sets called Global Named Sets.



- 7) Click Global Named Sets.
- 8) Click  button at the top right corner of the Named Sets pane.
- 9) Populate the form fields. **No spaces are allowed in the Name!**

Comment	All members of the dimension Customer and attribute Account
Name	AllAccounts
Definition	[Customer].[Account]

- 10) Click the check mark  at the top of the form. You should now have a Named Set called AllAccounts.

Repeat the steps from 8 to 10 to create other Named Sets as shown below:

Comment	All members of the dimension Product and attribute SKU
Name	AllProducts
Definition	[Product].[SKU]

Comment	All members of the dimension Time and attribute Month
Name	PlanningMonths
Definition	[Time].[Month]

Comment	Current Month
Name	CurrentMonth
Definition	&PlanningMonths.Filter(#[Month\$IsCurrent])

Comment	Future Months
Name	FutureMonths
Definition	Time.[Month].filter(#[Month\$InPast] == False && #[Month\$IsCurrent] == False)

Comment	Current and Future Months
Name	CurrentAndFutureMonths
Definition	&PlanningMonths.Filter(#.Key>=&CurrentMonth.element(0).Key)

Comment	Current Year
Name	CurrentYear
Definition	&PlanningMonths.Filter(#.[Month\$IsCurrent]).relatedmembers([Year])

Comment	Current working version
Name	CWV
Definition	[Version].[Version Name].[CurrentWorkingView]

Comment	CWV and Scenarios
Name	CWVAndScenarios
Definition	[Version].[Version Name].filter(#.Key <= 0)

Comment	Last Month
Name	LastMonth
Definition	&PlanningMonths.Filter(#.[Month\$IsCurrent]).element(0).leadoffset(-1)

Comment	Next Month
Name	NextMonth
Definition	&PlanningMonths.Filter(#.[Month\$IsCurrent]).element(0).leadoffset(1)

Comment	Past Months
Name	PastMonths
Definition	&PlanningMonths.filter(#.[Month\$InPast] == True)

Comment	All members of the dimension Time and attribute Quarter
Name	PlanningQuarters
Definition	[Time].[Quarter]

Comment	All members of the dimension Time and attribute Year
Name	PlanningYears
Definition	[Time].[Year]

Rules & Scope

A Rule can be a formula, a calculation, or anything that can impose certain business conditions and restrictions as well. All Rules are always bound by a scope, for every version.

A Scope determines the boundary or limitations to apply rules. It constitutes a scope statement that specifies all the grains, where a Rule is applied. If any grain is skipped data will be aggregated and formula is calculated

Rules are written as part of scope statements if the values of measures (used in the rule) remain mostly static.

There are many Scope Types available in the o9 platform. A few important ones are listed below:

Regular	For general rule calculations
Cartesian	Used primarily to insert intersections
Recurrence	Used for loop calculations (on-hand)
Block	Used to execute rules in sequence
Procedure	Used to define scopes for executing procedures
Spread	Used for data disaggregation

Write and Execute Regular Scope Statements for the below:

- 1) Calculate the difference between Shipment and Consumption values and store them in a new measure called Delta.
- 2) Assign an absolute value to the above measure for a given selection of Product, Customer, Month (Hint: Use the function Abs)
- 3) Create a Measure to Calculate Ratio between Channel Inv. and Consumption and write a rule to calculate the same?
- 4) Write a Regular scope to calculate a few new measures, such that each new measure holds 10% value of Each of the Budget, Channel Inventory Shipment, Forecast & Consumption measures.
- 5) Make the Budget for Product Group XYZ and Customer ABC, 1.5 times its original value.
- 6) All the historical data has to be actualized. For all Past Months,
 - a. Copy the Forecast values into a new measure called FcstActuals.
 - b. Create a ShipActuals measure and copy the shipments into this.
- 7) The Yearly aggregated value of Forecast & Shipment Actuals has to be calculated. Create new measures at higher level and write rules to store the aggregated values.
- 8) You need to capture the Last Cycle (Month) Forecast values. Create a measure called LC Forecast, and write a rule to capture the Last cycle forecast values, for the last 3 months from Current Month.

Cartesian Scope

Story 1:

Candy and More is on an expansion spree and is planning to open new customer distribution centers in untapped regions. Walmart has been generating good revenue in the American region and now prepared to enter into the Indian market.

Objective

To learn: About the difference between Regular Scope vs Cartesian Scope; Assignment of a measure with a constant; How to read logs for invocations, executions? How to create Fact File for data load? About ways to create intersections (via IBPL and/or Fact File).

Info: “Customer – Region Association” is an association measure to relate the Customers to their operational regions grained at Account – Country (lowest level in each dimension). Eg: If Kroger is operational in USA then Measure.[Customer – Region Association] = 1 @ Kroger – USA intersection.

Inserting New Intersections (Measure data)

Can be done in two ways – upload fact file or use cartesian scope

Step1 - Create the below Measures and Measure Groups

- Create a Measure Group named AccountCountryAssociation
- Create two Measures –
 - CustomerRegionAssociation
 - RegionalBudgetAllocation%
- Granularity – Account and Country

Name	CustomerRegionAssociation
Description	Association measure to relate the Customers to their operational regions
Data Type	Number
Type	Regular
Format	#,##0;(#,##0)
Aggregation	Null

Associate Walmart with India using IBPL rule via cartesian scope statement.

Cartesian scope: (&AllCountries.[India] * &AllAccounts.[Walmart] * &CWV);

Measure.[CustomerRegionAssociation] =1;

End scope;

Another method is to upload a fact life that has the Customer Region Association values. To create the remaining intersections in the platform, upload the Fact.Regional Allocation file using the Data Management workspace.

Spread Scope

Story 2

Candy and More has two planning levels:

Strategic Planners (COOs): Responsible for allocation of the annual operational budget to Customer Accounts and review/approve the regional requirements submitted by the Regional Planners.

Regional Planners: Responsible for analyzing and submitting the budget requirements for the upcoming year to the Strategic Planners for review/approval. Based on the approved regional allocation percentages, generate the plan for each region in which the customer account is operational.

Objective

To learn: How to work with multiple measure groups on the o9 platform

Infer the required grain of following measures and create them on mPower/o9 Platform.

Step2 - Create the below Measures and Measure Groups

- ✓ Create a measure called “RegionalBudget” under RegionalPlan MG (this takes a union of Budget and RegionalBudgetAllocation% grains and is calculated)
- ✓ Budget is at Account/Product/Month AND RegionalBudgetAllocation% is at Country/Account grains, hence RegionalBudget should be at Account/SKU/Month/Country grain
- ✓ So, for the RegionalPlan MG, set the granularity as SKU, Country, Month, Account

Step3 - select to see the aggregated Budget data for Walmart before calculating the Regional Budget
select ([Customer].[Account].[Walmart] * [Version].[Version Name].[CurrentWorkingView]) on row,
({Measure.[Budget]}) on column;

Step4 - select to see the allocation % before calculating the rule for Regional Budget for Walmart
select ([Customer].[Account].[Walmart] * [Region].[Country] * [Version].[Version Name].[CurrentWorkingView]) on row,
({Measure.[CustomerRegionAssociation], Measure.[RegionalBudgetAllocation%]}) on column;

Step5 - select to view the Regional Budget that will be empty

```
select ( [Customer].[Account].[Walmart] * [Region].[Country] * [Version].[Version Name].[CurrentWorkingView] ) on row,
({Measure.[RegionalBudget]}) on column;
```

Step6 - Write the scope statement to calculate the allocated Regional Budget.

Objective	Calculating Regional Budget for Walmart
Scope statement	scope: ([Customer].[Account].[Walmart] * [Region].[Country] * [Product].[SKU] * [Time].[Month] * [Version].[Version Name].[CurrentWorkingView]); Measure.[RegionalBudget]= if (Measure.[CustomerRegionAssociation]==1) then Measure.[Budget] * Measure.[RegionalBudgetAllocation%] else Null; end scope;

Objective

To understand and visualize the concept of Disaggregation. To learn the difference between Regular Scope vs Spread Scope. How to write a Spread Scope?

Info: Based on his experience and looking at the past data, Mr. X (one of the COOs at Candy and More who handle Kroger's account) wishes to cap the annual budget at Year level.

Design the solution such that when Mr. X enters the Budget at year level for Kroger, it is proportionately distributed across all the months for each product.

(Hint: Think of measure spreading)

Set the Measures for Spreading using the Model Designer- (to be done for Budget and Regional Budget)

- Select the Basis Measure Type
- Select the Basis Measure
- Select the Spreading type

Step7 – Writing a Spread Scope statement for Updating the Budget value

```
Spread scope: ([customer].[Account].[Walmart] * [Version].[Version Name].[CurrentWorkingView]);
Measure.[Budget]=900000;
End scope;
```

Step8 - Select to see the existing Regional Budget value-

```
select ([Customer].[Account].[Walmart] * Region.[Country] * [Version].[Version Name].[CurrentWorkingView] ) on row,
({Measure.[RegionalBudget]}) on column; // this will be based on the old Budget value
(Since Budget values are at month level, Regional Budget needs to be spread across all Countries from Month level) using spread scope –
```

Step9 - Spreading the Regional Budget for Walmart (can be set as an active rule)–

spread scope: ([Customer].[Account].[Walmart]
* [Version].[Version Name].[CurrentWorkingView]);
Measure.[RegionalBudget] = Measure.[Budget];
end scope;

Step10 - check the Regional Budget amount now and see whether the correct % has been allocated based on the new Budget value–

select ([Customer].[Account].[Walmart] * [Region].[Country] * [Version].[Version
Name].[CurrentWorkingView]) on row,
({Measure.[Regional Budget]}) on column;

Exercises:

- 1) Copy Measure.XYZ (Granularity- Item Group x Location x Month) to Measure.ABC (Granularity- Item x Location x Month) such that the Items of that Item Group inherits the same value.
- 2) Copy Measure.XYZ (Granularity- Item Group x Location x Month) to Measure.ABC (Granularity- Item x Location x Month) such that the value is equally distributed to items of that Item Group.
- 3) Copy Measure.XYZ (Granularity- Item Group x Location x Month) to Measure.ABC (Granularity- Item x Location x Month) such that the value is equally distributed to Items of that Item Group but only where Measure.DEF (Granularity- Item x Location Month) is set to 1 (this measure resides in the same measure group)

Recurrence Scope

Story 3

Inventory Manager at Candy and More is responsible to keep track of inventory month over month and maintain the inventory levels to avoid any stock outs. The inventory at the customer DC gets refreshed each month based on the following formula:

Channel Inv (Current Month) = Channel Inventory (Last Month) + Shipments – Consumption

Objective

To understand the usability and need of Recurrence Scope. To learn the difference between Regular Scope vs Recurrence Scope. How to write a Recurrence Scope?

Ensure that the Measure Channel Inv exists under Planning Measures or else Create the measure “Channel Inv” on mPower.

Channel Inv: Channel Inventory is the value of the SKUs present at a customer distribution center at any point in time.

Write a rule to calculate Channel Inv for Feb 2019 to CM-1 (i.e. last month).

Recurrence Scope	&AllProducts * &AllAccounts * &PastMonths * &CWV
Formula	Measure.[Channel Inv] = if (Time.#.Key == &PastMonths.element(0).Key) then Measure.[Channel Inv] else coalesce(Measure.[Channel

	Inv]@(Time.#.LeadOffset(-1)),0) + coalesce(Measure.[Shipment],0) - coalesce(Measure.[Consumption],0);
--	---

Active Rules

Story 4:

To be financially prepared, Candy and More wants to have visibility to the sales outlook since its foundation and calculate metrics to depict the year over year trends and variation from the annual planned Budget.

Objective

To learn: How to write a rule? How to define the scope? How to use Filters in scope statement? How to use Named-sets in scope? How to use relatedmembers? How to write 'if' condition? How to use logical operators? How to use Coalesce, Safedivide, IsNull, Leadoffset? How to refer scope member in the rule?



What should be the aggregation type of any measure so that at any level it gives the ratio between two numbers?

Ans: It should be computed type with formula

Create the below Measures created under the PlanningMeasures MG

Name	Outlook	Outlook vs Budget Gap	Outlook vs Budget %
Description	Historic shipments and forecasted demand	Difference between Outlook and Budget	Percentage of Outlook w.r.t Budget
Data Type	Number	Number	Number
Type	Regular	Regular	Regular
Format	#,##0;(#,##0)	#,##0;(#,##0)	0.0%
Aggregation	Sum	Sum	Computed
Aggregation Formula	-	-	if(Measure.[Budget]!=0) then Measure.[Outlook] / Measure.[Budget]

Write rules to calculate the Outlook Measure for different time buckets.

Scope	&AllProducts * &AllAccounts * &PastMonths * &CWV
Formula	Measure.[Outlook]=Measure.[Shipment];
Scope	&AllProducts * &AllAccounts * &CurrentMonth * &CWV
Formula	Measure.[Outlook]=coalesce(Measure.[Forecast],5000);
Scope	&AllProducts * &AllAccounts * &FutureMonths * &CWV
Formula	Measure.[Outlook]=coalesce(Measure.[Forecast],1000);

Scope	&AllProducts * &AllAccounts * &PlanningMonths * &CWV
Formula	Measure.[Outlook vs Budget Gap] = coalesce(Measure.[Outlook],0)-coalesce(Measure.[Budget],0);
Formula	Measure.[Outlook vs Budget %] = Measure.[Outlook]/Measure.[Budget];

Alternatively, you can write a rule to populate “Outlook vs Budget %” only if Outlook vs Budget % > 30%.

Scope	&AllProducts * &AllAccounts * &PlanningMonths * &CWV
Formula	Measure.[Outlook vs Budget %] = if(Measure.[Budget] != 0 && Measure.[Outlook]/Measure.[Budget]>0.30) then Measure.[Outlook]/Measure.[Budget] else Null;

Also create the below LY measures under the same Measure Group

Name	LY Outlook	Outlook vs LY Outlook Gap	Outlook vs LY Outlook %	Outlook vs LY Outlook Gap %
Description	Value of Outlook for same time last year	Difference between Outlook and LY Outlook	% of Outlook compared to Last Year	% of the Outlook Gap compared with Last Year's value
Data Type	Number	Number	Number	Number
Type	Regular	Regular	Regular	Regular
Format	#,##0;(#,##0)	#,##0;(#,##0)	0.0%	0.0%

Aggregation	Sum	Sum	Computed	Computed
Aggregation Formula			if(Measure.[LY Outlook]!=0) then Measure.[Outlook] / Measure.[LY Outlook]	if(Measure.[LY Outlook]!=0) then Measure.[Outlook vs LY Outlook Gap] / Measure.[LY Outlook]

Name	LY Consumption	Consumption vs LY Consumption Gap	Consumption vs LY Consumption %	Consumption vs LY Consumption Gap %
Description	Units Consumed at the Customer DC same time Last Year	Difference between Consumption and LY Consumption	% of Consumption compared to Last Year	% of the Consumption Gap compared with Last Year's value
Data Type	Number	Number	Number	Number
Type	Regular	Regular	Regular	Regular
Format	#,##0;(#,##0)	#,##0;(#,##0)	0.0%	0.0%
Aggregation	Sum	Sum	Computed	Computed
Aggregation Formula			if(Measure.[LY Consumption]!=0) then Measure.[Consumption] / Measure.[LY Consumption]	safedivide(Measure.[Consumption vs LY Consumption Gap],Measure.[LY Consumption])

Write rules to calculate the measures LY Outlook and LY Consumption.

Scope	&AllProducts * &AllAccounts * &PlanningMonths * &CWV
Formula	Measure.[LY Outlook] = Measure.[Outlook]@(Time.#.Leadoffset(-12));
Formula	Measure.[LY Consumption] = Measure.[Consumption]@(Time.#.Leadoffset(-12));

Write rules to calculate Gap, % and Gap % between Outlook and LY Outlook.

Scope	&AllProducts * &AllAccounts * &PlanningMonths * &CWV
Formula	Measure.[Outlook vs LY Outlook Gap] = coalesce(Measure.[Outlook],0) - coalesce(Measure.[LY Outlook],0);

Formula	Measure.[Outlook vs LY Outlook %] = SafeDivide(Measure.[Outlook],Measure.[LY Outlook],0);
Formula	Measure.[<i>Outlook vs LY Outlook Gap %</i>] = Measure.[<i>Outlook vs LY Outlook %</i>] - 1;

Write rules to calculate Gap and % between Consumption and LY Consumption.

Scope	&AllProducts * &AllAccounts * &PlanningMonths * &CWV
Formula	Measure.[Consumption vs LY Consumption <i>Gap</i>] = coalesce(Measure.[Consumption],0) - coalesce(Measure.[LY Consumption],0);
Formula	Measure.[Consumption vs LY Consumption %] = SafeDivide(Measure.[Consumption],Measure.[LY Consumption],0);
Formula	Measure.[Consumption vs LY Consumption Gap %] = Measure.[Consumption vs LY Consumption %] - 1;

Widgets

The different types of widgets available on o9 platform are Action Buttons, Bubble Chart, Calendar, Chart, External Service Report, Geo Map, Indented Grid, KPI, Master Data, NLP Report, Network, Pivot, Pulse, Scenario Manager, Task, and Where Used.

We will be creating Pivot and Chart widgets to display the data in this exercise.

Line Chart

- 1) Open the **Designer** workspace.
- 2) Ensure you are in **Report Designer**, if not, click **Edit Reports** button to switch from Model Designer to Report Designer.



- 3) Click **Chart** button to create a Line/Bar chart.
- 4) Type in the name **Outlook** for the widget in the text box 'Report Name'.
- 5) From the 'Dimension' pane drag the level attributes and drop them on to the **Filter** pane. In this case the level attributes are: Version Name, All Accounts, Channel, Account, All Products, Product Category, and SKU. This is used to filter the widget based on values in the level attributes. **NOTE: You can alternatively click on the funnel icon (Add to Filters) for each level attribute, rather than dragging them to the Filter pane.**
- 6) **Adding Version Name attribute in the widget filters, is mandatory.**
- 7) From the 'Dimension' pane drag the level attributes and drop them on to the **X-Axis Levels**. In this case the level attributes are: Year, YTD, Quarter, and Month. **NOTE: You can alternatively click on the arrow-right icon (Add to X-Axis Levels) beside each level.**

- 8) Click the gear icon for each level attribute in the X-Axis Levels pane and populate fields as below: Click the check mark icon to Save after completing the settings for each level attribute.x

Year:

Short Display Name	Y
Is Default	No

YTD:

Short Display Name	YTD
Is Default	No

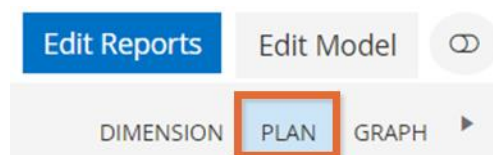
Quarter:

Short Display Name	Q
Is Default	No

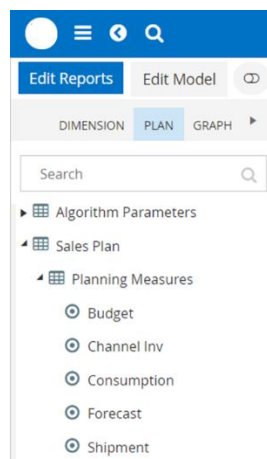
Month:

Short Display Name	M
Is Default	Yes (check the checkbox)

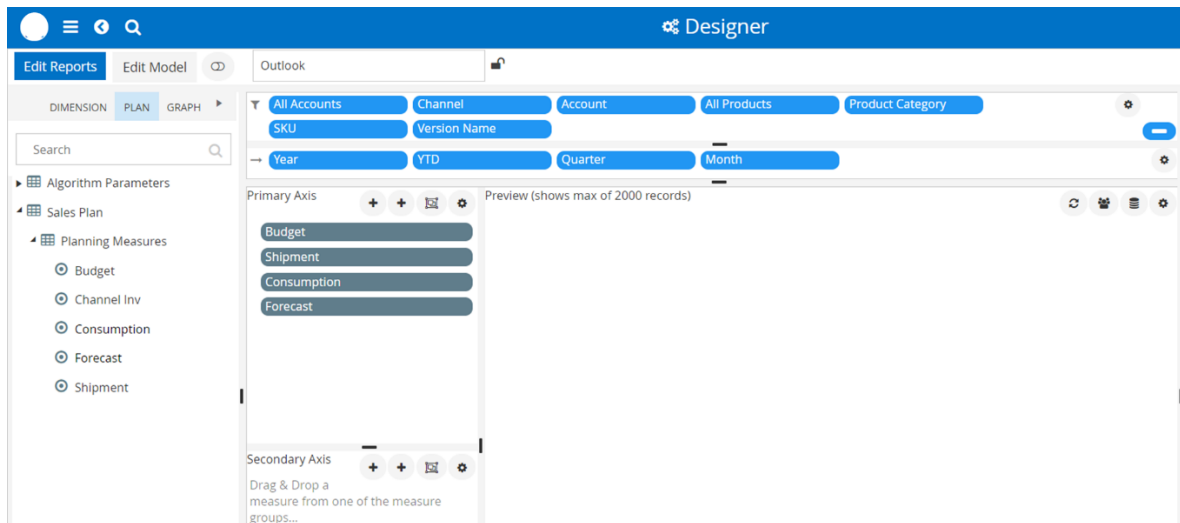
- 9) Click the **Plan** tab.



- 10) Click the arrow head next to Plan- 'Sales Plan' and then click the arrow head next to measure group – 'Planning measures'.

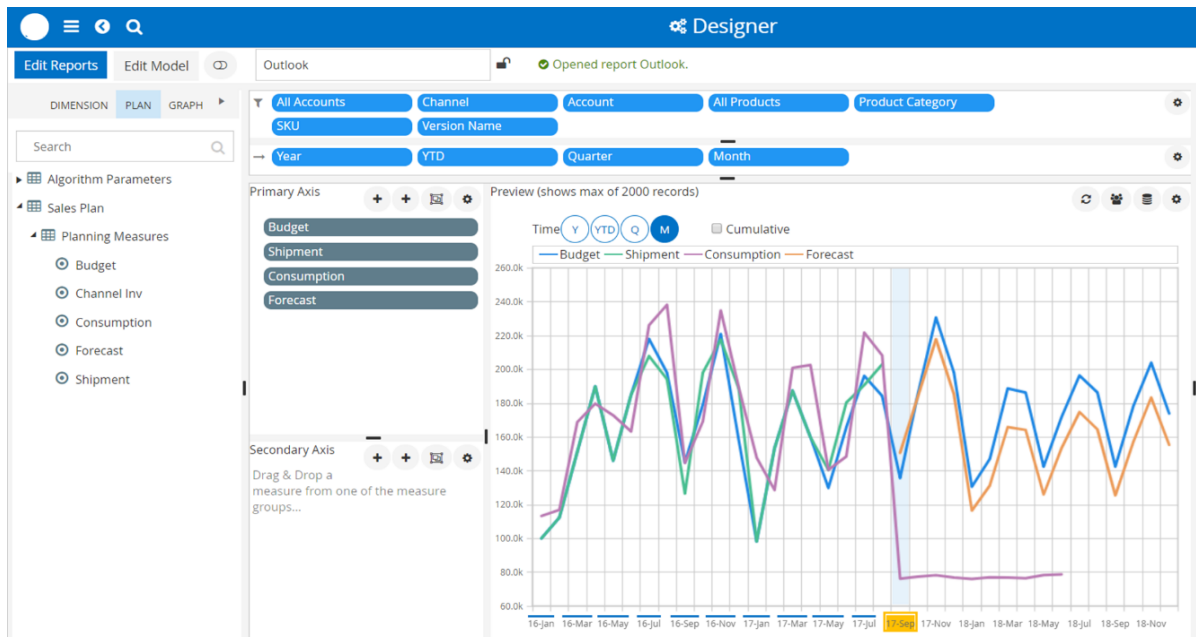


- 11) Drag the measures and drop them on to the **Primary Axis** pane. In this case the measures are: Budget, Shipment, Consumption, and Forecast. Alternatively, one can also populate the measures in the Primary Axis pane is by clicking the first + icon and typing in the measures, or click on the first + icon beside each measure name.



12) Click **Preview** button to display the line chart.

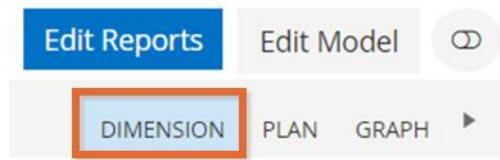
13) Click **Save** button.



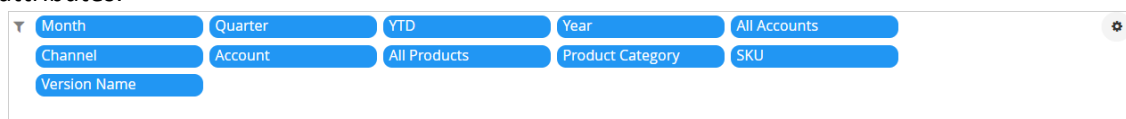
Pivot Table

- 1) Open the **Designer** workspace.
- 2) Ensure you are in **Report Designer**, if not, click **Edit Reports** button to switch from Model Designer to Report Designer.

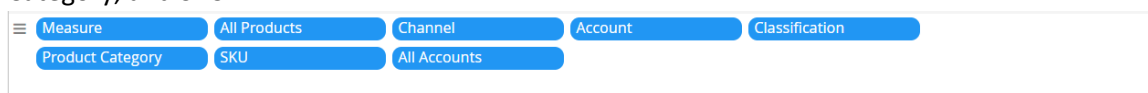
- 3) Click **Pivot** button to create a Pivot widget.
- 4) Type in the name 'Breakdown' for the widget in the text box 'Report Name'.
- 5) Click the **Dimension** pane.



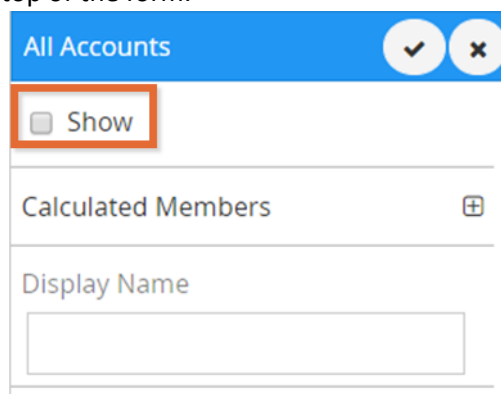
- 6) Drag the level attributes and drop them on to the **Filter** pane. In this case the level attributes are: Version Name, Month, Quarter, YTD, Year, All Accounts, Channel, Account, All Products, Product Category, and SKU. This is used to filter the widget based on values in the level attributes.



- 7) From the 'Dimension' pane drag the level attributes and drop them on to the **Row** pane. In this case the level attributes are: All Accounts, Channel, Account, All Products, Classification, Product Category, and SKU.

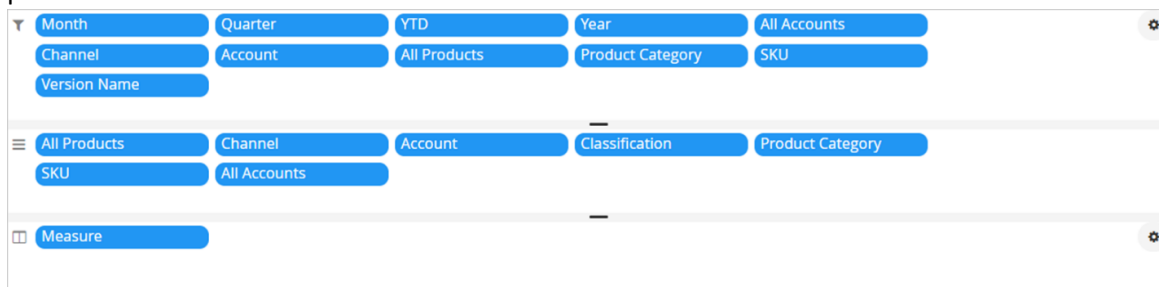


- 8) Click the gear icon for All Accounts.
 9) Uncheck 'Show'
 10) Click the button at the top of the form.

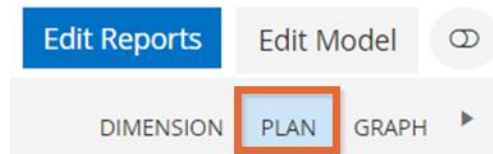


Repeat steps 8 to 10 for Channel, All Products, Classification, and Product Category.

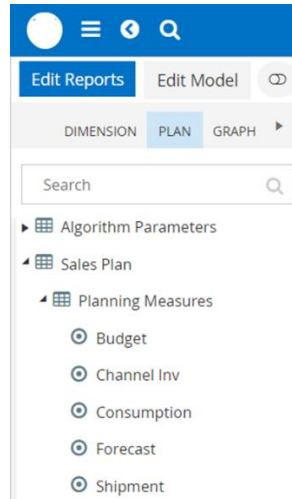
- 11) Drag the 'Measure' pill from Row pane and drop on to the Column pane.



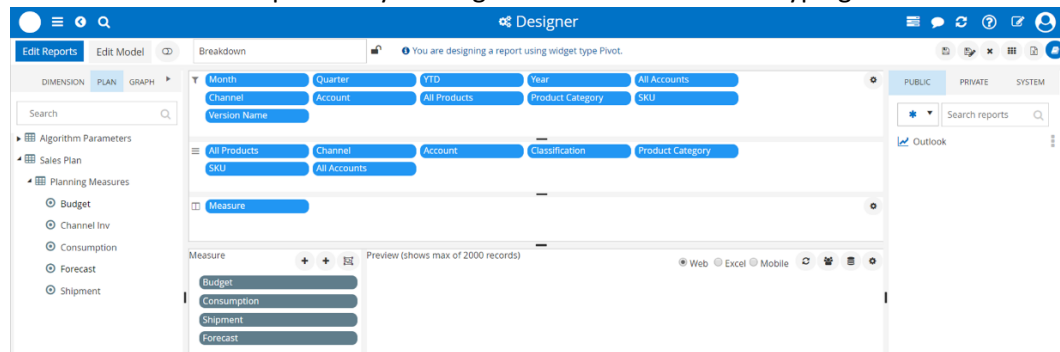
- 12) Click the **Plan** tab.



- 13) Click the arrow head ► next to Plan- 'Sales Plan' and then click the arrow head ► next to measure group – 'Planning measures'.



- 14) Drag the measures and drop them on to the **Measure** pane. In this case the measures are: Budget, Shipment, Consumption, and Forecast. Alternatively, one can also populate the measures in the Measure pane is by clicking the first + button and typing in the measures.



- 15) Click **Preview** button to display the Pivot widget.

- 16) Click **Save** button.

Designer

Breakdown

Opened report Breakdown.

Month Quarter YTD Year All Accounts Channel Account

All Products Channel Account Classification Product Category SKU All Accounts

Measure

Preview (shows max of 2000 records)

Showing 24 rows of data.

Account	SKU	Budget	Consumption	Shipment	Forecast
Kroger	Butterfinger	213,603	155,612	116,525	101,136
	Cerelac	96,323	75,880	54,614	34,179
	Crunch	160,657	205,424	80,300	71,882
	Gerber	149,610	113,125	96,142	55,095
	Gerber Graduates	109,348	89,690	68,004	47,998
	KitKat	264,346	202,514	143,064	134,088
	Milo	84,152	48,157	44,468	40,150
	Nestea	93,261	43,593	41,756	46,060
SAMS	Butterfinger	444,420	209,703	228,880	176,543
	Cerelac	345,981	197,763	212,553	125,467
	Crunch	208,793	103,032	109,880	94,859
	Gerber	209,531	99,231	108,181	88,948
	Gerber Graduates	180,482	104,145	116,736	71,387
	KitKat	194,703	100,290	105,520	72,429
	Milo	159,021	105,895	115,567	51,671

Note: Under the measures pane of the Pivot, one can create a new Group by clicking the Create Group button . Once a group is created, drag and drop the measures under respective groups.

Designer

Breakdown Report 2

Opened report Breakdown Report 2.

Month Product Category SKU All Accounts Account Version Name

Month Account Product Category SKU

Measure

Preview (shows max of 2000 records)

Showing 329 rows of data.

Month	Account	Product Category	SKU	Budget	Shipment	Consumption	Forecast	Outlook vs Budget	Outlook vs BudgetGap	Outlook vs Budget %
18-Jan	Walmart	Baby Foods	Cerelac	1,847	6,189	6,598		335.1%		
		Gerber Gradu..	Butterfinger	2,855	9,565	8,766		335.1%		
		Chocolate	Crunch	3,190	10,689	10,798		335.1%		
			KitKat	2,920	9,784	8,982		335.1%		
		Refreshment	Milo	3,056	10,239	9,836		335.1%		
			Nestea	3,510	11,761	11,464		335.1%		
				4,557	15,271	15,696		335.1%		
18-Feb	Walmart	Baby Foods	Cerelac	1,944	6,513	5,909		335.1%		
			Gerber	9	30	20		335.1%		
		Chocolate	Gerber Gradu..	2,020	6,768	9,729		335.1%		
			Butterfinger	3,259	10,920	10,949		335.1%		

Workspaces

Workspaces

- 1) Click the **Design Layout** icon at the top right of the window to enter design mode.
- 2) Expand the Workspace browser by clicking the **Workspace** button at the top left of the window.
- 3) Click the button at the top of the Workspace browser.
- 4) Enter the following values in the Workspace form:

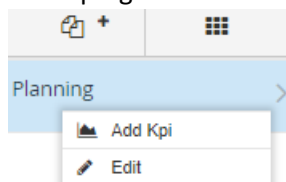
Title	Planning
Icon	plane
Hidden?	FALSE (leave unchecked)
IsMobile	FALSE (leave unchecked)
Roles	public
Default page title	Planning
Default view title	Planning

- 5) Click the button at the top of the form.

Pages and Page Groups

The Workspace is created and labeled Planning. You are now going to edit the existing page to rename it and place it in a page group.

- 1) Hover over the Planning tab and click the 3 vertical dots. Select Edit. (If you do not see the three dots on the New Page tab, you may not be in Design Mode. Click the Design Layout button at the top right of the window to enable Design Mode.)

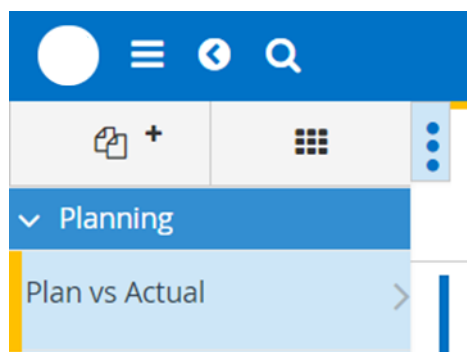


- 2) Enter these values in the Page form:

Title	Plan vs Actual
IsDefault	TRUE (leave checked)
Show View Header	TRUE (leave checked)
Page Group	Planning (type the word into the field)

- 3) Click the button at the top of the form.

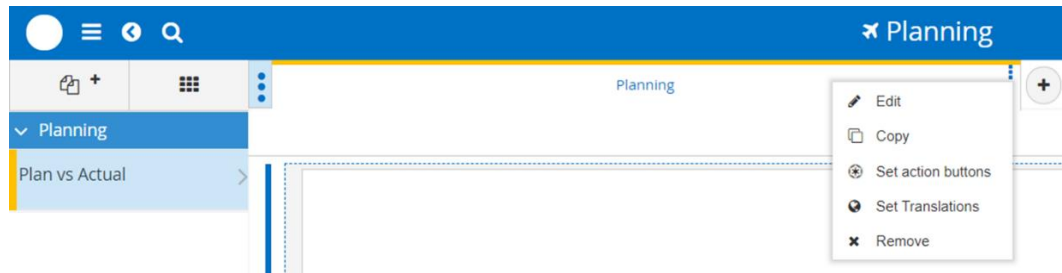
There will now be a page group labeled Planning with a single page in it labeled Plan vs Actual.



Views

The page you created has a default view labeled Planning. You will now edit this view.

- 1) Hover over the new view tab (Planning, in this case) and click the 3 vertical dots. Select **Edit**. (If you do not see the three dots on the New Page tab, you may not be in Design Mode. Click the **Design Layout** button at the top right of the window to enable Design Mode.)



2) Enter the following values in the form:

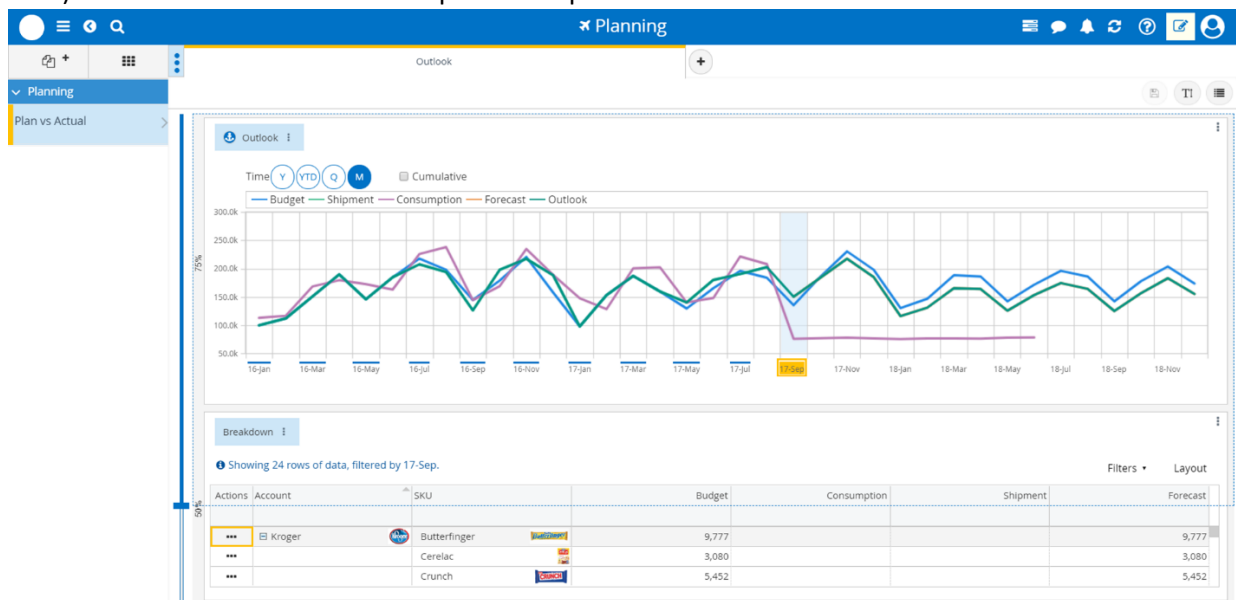
Title	Outlook
Has Anchored Report	TRUE (leave checked)
Compact View	FALSE (leave unchecked)
IsDefault	TRUE (leave checked)
Roles	public
Config	Two Vertical stacked

3) Click the button at the top of the form.

Adding Widgets

Now you will add the widgets we have already created to this form.

- 1) Click the **Open Reports List** icon at the top right of the Outlook view.
- 2) Drag the Outlook report from the Reports List to the top pane of the view.
- 3) Drag the Breakdown report from the Reports List to the lower pane.
- 4) Click the button at the top of the Reports List to close it.

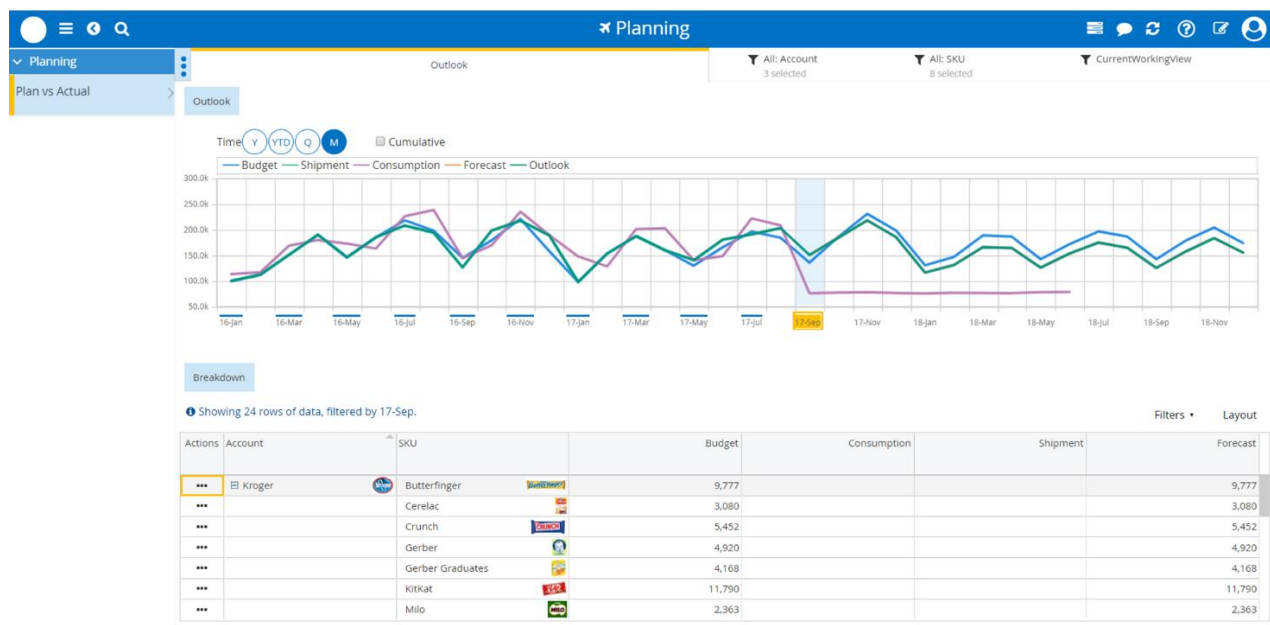


Setting Block Height

You have added the report widgets to your view. Now you will adjust the height of the two widget blocks to show them both in the same window.

- 1) Click the **Set Blocks Height** button near the top right of the window.
- 2) Set the First Row to 50% and the Second Row to 50%.
- 3) Click the button to close the form.
- 4) Click the **Design Layout** button to exit design mode.

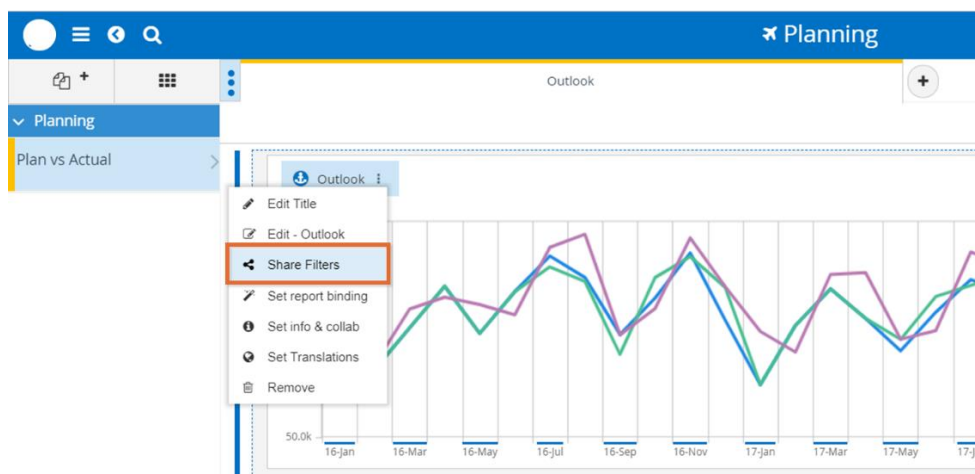
Your view should now look something like this:



Filtering (Sharing and Linking)

It is possible to link the filters of the chart and pivot table, so that you can navigate the data in the pivot table by interacting with the chart.

- 1) Open the **Workspace Navigator** .
- 2) Select the **Planning** workspace.
- 3) Select the **Plan vs Actual** page.
- 4) Click the **Design Layout** icon.
- 5) Click the three dots on the **Outlook** tab and select **Share Filters**.



- 6) Select Scope as **View** for **Version Name**, **Account**, and **SKU**.
- 7) Click the  button to close the form.
- 8) You can select filter sharing scope as View/Page/Workspace/Global. Accordingly, the filter will be shared within the view or across the page or across the workspace.
- 9) Click the three buttons on the Breakdown tab and select **Link Filters**.
- 10) Fill in the form with these values:

- 11) Click the  button to close the form.

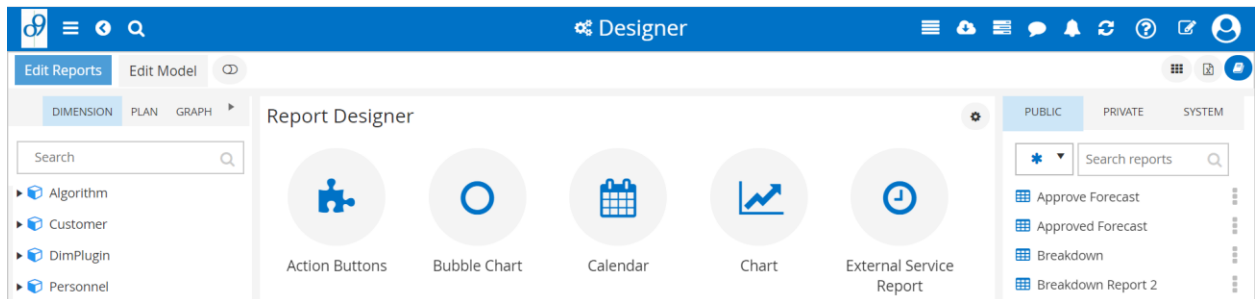
You can now click the date values along the X axis of the chart to filter the contents of the pivot table.

Member Filter:

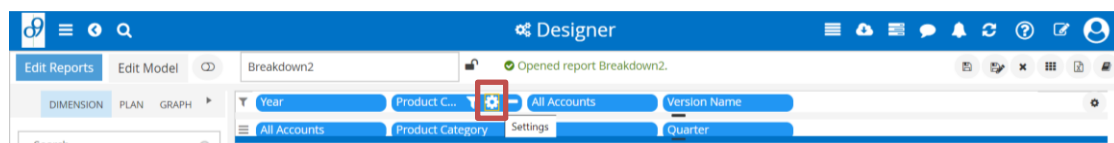
Member filters restrict the data display in report, to specific members of one or more level attributes.
Ex: Display the data of only the 'Chocolate' Product Category.

Configuring member filter in a Pivot widget:

- 1) Open the **Designer** workspace.
- 2) Ensure you are in **Report Designer**, if not, click **Edit Reports** button to switch from Model Designer to Report Designer.
- 3) Click **Widget Library**  button to open the list of reports created.
- 4) Select the 'Breakdown' pivot widget from the reports list.



- 5) Edit the report name to 'Breakdown2' and click the Save As icon. This will create a copy of the 'Breakdown' pivot.
- 6) From the **filter** pane of the report, select the member for which filter is to be applied. In this case, click the Gear icon for the member 'Product Category'.



- 7) The member property page for 'Product Category' will open on the left side. In the Member Filter field, write the condition to set the filter for the category Chocolate.

Edit Reports
Edit Model

Product Category

Dimension Alias Name

Alias Name

☐ Single Select

☐ Currency Filter

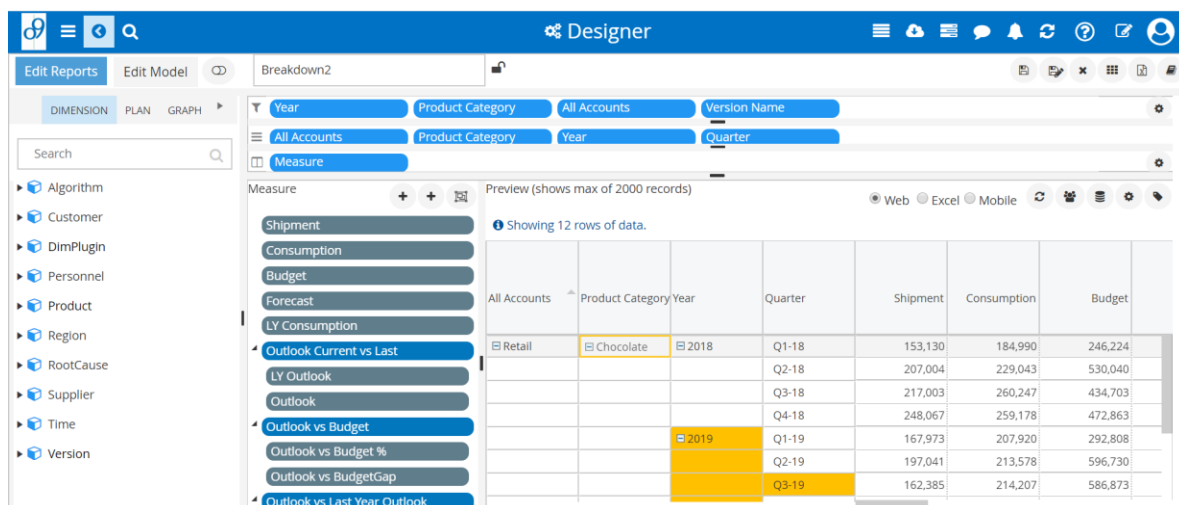
Member Filter

[Product].[Product Category]
.[Chocolate]

Default Value Expression

Enter Condition...

- 8) Click the button at the top of the form.
- 9) Click **Preview** button to refresh the data in Pivot widget. You should see the report data filtered against the category Chocolate.



10) Click **Save** .

Measure Filter:

Measure filters restrict display of measure data in the report, based on specific measure conditions. Ex: Display the records where Sales value > 100000.

They can be setup in the Data Properties window of the widget.

Configuring measure filter in a Pivot widget:

- 1) Open the **Designer** workspace.
- 2) Ensure you are in **Report Designer**, if not, click **Edit Reports** button to switch from Model Designer to Report Designer.
- 3) Click **Widget Library** button to open the list of reports created.
- 4) Select the 'Breakdown2' pivot widget from the reports list.
- 5) Click the **Data Properties** icon in the right-side corner of the widget screen. The Data Properties pane will open on the left.
- 6) In the Measure Filter field, add the condition to filter the data points which have Budget Measure higher than 300000.

Edit Reports
Edit Model

Data

☐ Show Inactive Members

☐ Include Nulls

☐ Nulls For Finer Grain Select

☐ Nulls For Unrelated Attr Select

☐ Require Edit Validation

Sub Totals Type

NoSubtotals

Include Nulls For Series

Measure Filter

Measure. [Budget] > 300000

- 7) Click the button at the top of the form.
- 8) Click **Preview** button to refresh the Pivot widget. You should see the report data with Budget value > 300000.

Designer

Breakdown2

Data

☐ Show Inactive Members

☐ Include Nulls

☐ Nulls For Finer Grain Select

☐ Nulls For Unrelated Attr Select

☐ Require Edit Validation

Sub Totals Type

NoSubtotals

Include Nulls For Series

Measure Filter

Measure. [Budget] > 300000

Measure

Shipment

Consumption

Budget

Forecast

LY Consumption

Outlook Current vs Last

LY Outlook

Outlook

Outlook vs Budget

Outlook vs Budget %

Outlook vs BudgetGap

Outlook vs Last Year Outlook

Outlook vs LY Outlook %

Preview (shows max of 2000 records)

Showing 10 rows of data.

All Accounts	Product Category	Year	Quarter	Shipment	Consumption	Budget
Retail	Chocolate	2018	Q2-18	207,004	229,043	530,040
			Q3-18	217,003	260,247	434,703
			Q4-18	248,067	259,178	472,863
		2019	Q2-19	197,041	213,578	596,730
			Q3-19	162,385	214,207	586,873
			Q4-19		78,093	664,004
		2020	Q1-20		77,465	500,649
			Q2-20		76,837	568,504

- 9) Click **Save** .

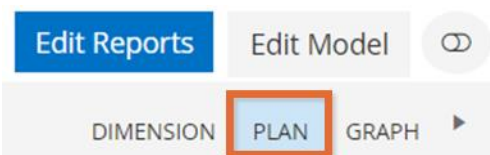
Conditional Formatting:

Conditional formatting can be applied on a measure to place a condition and differentiate the data range satisfying that condition in a specific color or with an indicator. This is always imposed on a measure at the Plan level and is applicable globally. However, in the widget properties window of the

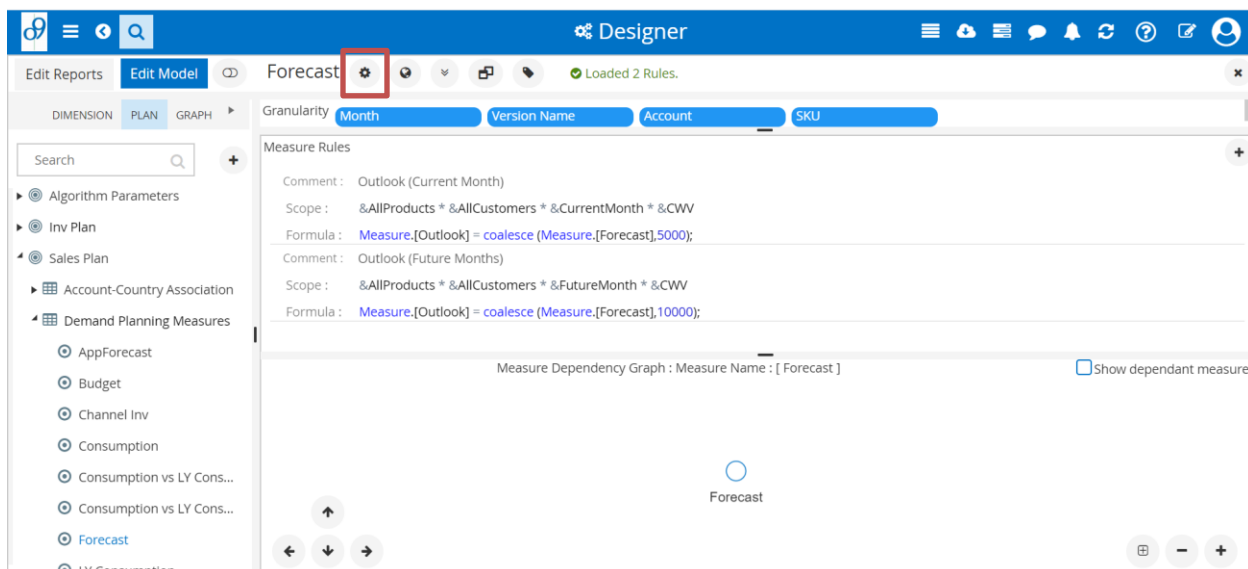
report, the Show Conditional Formatting option must be selected for the formatting to be visible in the report.

Configuring Conditional Formatting in a Pivot widget:

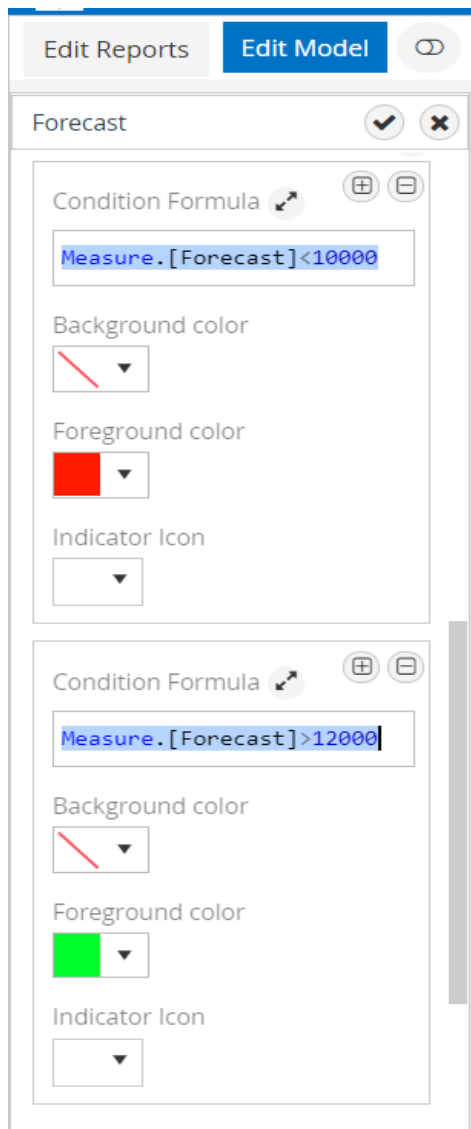
- 1) Open the **Designer** workspace.
- 2) Ensure you are in **Model Designer**, if not, click **Edit Model** button to switch from Report Designer to Model Designer.
- 3) Click the **Plan** tab.



- 4) Click the arrow head ► next to Plan- 'Sales Plan' and then click the arrow head ► next to measure group – 'Planning measures'. From the list of measures under this measure group, select Forecast.
- 5) Click the Gear ⚙ icon in the Forecast measure window



- 6) The Edit Measure window for the Forecast measure, opens on the left.
- 7) Click the + icon against the 'Conditional Formula in this window.
- 8) Write the desired condition, Measure.[Forecast]<10000, in the formula field. Also select the desired Foreground and Background colors.
- 9) Click the ⊕ icon against the above condition formula, to add another condition for the same Forecast measure.
- 10) Write another condition for the measure in the formula field, like Measure.[Forecast]>12000. Set the foreground and background colors appropriately.



- 11) Click the  button at the top of the form.
- 12) Click **Edit Reports** button to switch from Model Designer to Report Designer.
- 13) Click **Widget Library**  button to open the list of reports created.
- 14) Select the 'Breakdown2' pivot widget from the reports list.
- 15) Click the **Widget Properties**  icon in the right-side corner of the widget screen. The Widget Properties pane will open on the left.
- 16) Select the check-box 'Show Conditional Format'.

Widget

Title
Grid Title

☒ Editable

☒ Enable Expand Collapse

☒ Enable Column Filter

☒ Enable Column Sort

☒ Enable Column Reorder

☒ Enable Column Resize

☐ Enable Just-in-time Columns

☐ Enable Audit Trail

☐ Enable Column Auto-collapse

☐ Sparsify Row Selection

☒ Show Conditional Format

☒ Show Image

17) Click the button at the top of the form.

18) Click **Preview** button to refresh the Pivot widget. You should see the report data with the Forecast values formatted in the color set as per the appropriate condition.

Breakdown2
Opened report Breakdown2.

Year Product Category All Accounts Version Name

All Accounts Product Category Year Quarter

Measure

Measure Preview (shows max of 2000 records)

Showing 10 rows of data.

All Accounts	Product Category	Year	Quarter	Shipment	Consumption	Budget	Forecast	LY Consumption	LY Outlook	Outlook Current vs Last
Retail	Chocolate	2018	Q2-18	207,004	229,043	530,040				—
			Q3-18	217,003	260,247	434,703	17,000			—
			Q4-18	248,067	259,178	472,863	3,000			—
		2019	Q2-19	197,041	213,578	596,730	7,000	229,043	207,004	—
			Q3-19	162,385	214,207	586,873	61,248	260,247	217,003	—
			Q4-19	78,093	664,004	664,004	245,831	259,178	248,067	—
		2020	Q1-20		77,465	500,649	172,882	207,920	167,973	—
			Q2-20		76,837	568,504	187,353	213,578	197,041	—
			Q3-20			550,134	196,612	214,207	196,248	—
			Q4-20			604,786	283,710	78,093	245,831	—

19) Click **Save** .

Calculated Measures

Calculated Measures are specific to a Report or a Pivot. Their scope is limited only to the respective Report. Calculated Measure has a formula that you will need to specify while defining the same.

Configuring Calculated Measures

- 1) Open the **Designer** workspace.
- 2) Open any of the Pivot reports.
- 3) Under the Measure pane, click the second icon. If you hover over the same, you will see “Calculated Measure” written as a tooltip
- 4) A placeholder for formula will appear.
- 5) Specify any formula that you want to, such as Measure.[Shipment]-Measure.[Consumption] and click on the tick mark. You will see the Calculated Measure in a different color under the Measure pane.

Exercises:

- 1) Create a Pivot that displays Outlook, LY Outlook, Consumption, LY Consumption where the LY Outlook is greater than 15,000. Show the SKU, Category & Month. Choose any options as the filter
- 2) Create a Report that displays Quarter wise Shipment & Consumption values for the Year 2019.
- 3) Show Year-wise Budget, Shipment, Outlook & Forecast values where you can filter only by account
- 4) Display the Forecast value for every product & every account monthly for year 2020. Highlight the Forecast values less than 5000 in RED & greater than 5000 in GREEN
- 5) Create a chart where the Shipment, Consumption values are stacked for every SKU for every Year
- 6) Create the below in view screens
 - a. Create a workspace called Mgmt Reports and associate reports 2,3,5 within a single view
 - b. Create a page called Detailed Reporting & within this, have 2 views. Associate Reports 1 & 4 in each of these views
- 7) Create a Data Management workspace, that holds so many pages, equal to the number of dimensions in your model. Group all these pages into a Master Data page group. Each page should have a view that should consist the master data of respective dimension.
- 8) Create a Pivot which shows the Budget, Budget override, Forecast, Planner Override, Outlook, LY Outlook, Shipment and Consumption, for all the quarters of the years 2019 and 2020. Display them for all customers and Product Categories. The layout should have only the Time related attributes in the columns. Dynamics like filters, measure creation or measure edits have to be decided appropriately.
- 9) Create a Bar chart, that displays the Year, Quarter and Month details on the X-axis, and has the below measures stacked one above the other:
 - a. Outlook and LY Outlook
 - b. Shipment and Consumption
 Choose filters appropriately and Pull the reports 8 & 9 into a single view.

- 10) The Head of Demand Strategy should see Forecast, Shipment and Budget Data for all the Products. His/her dashboard should comprise the below:
- Quarterly KPI for Forecast, Shipment and Budget, which can be filtered by Product Categories
 - A bubble chart which shows the Outlook, Budget and the % variation between the two, in the 3-axes of the chart, for all the Product Categories
 - The KPI data should also be indicated in a pivot, displaying data for all products and all months
 - For all the products, show the Forecast of all the past months and corresponding shipment. Indicate using arrows and colours to distinguish the intersections where Forecast is greater than or less than the corresponding Shipment value.
 - A Bar Chart that shows the Forecast and Shipment trends across Month, Quarter and Year. Choose the filters as necessary. Create a measure on the fly, that computes the Gap between the above two measures and shows the trend as a line.
 - All the above 5 reports should be part of the Demand Executive workspace. The Page should have one view, split into 4 widget panes. *Note: Choose your filters for all the reports in such a way that they display data synchronously in the dashboard.*

Action Buttons

Action buttons enable user driven actions. They give the control or power to the end users to perform certain actions. The main difference between an action button and an active rule is that an action button can trigger an action or a procedure, on-demand.

An action button can be setup globally or can be confined to specific reports. Also, you can restrict specific user/role access for the action buttons.

An Action button is a widget in the Report Designer workspace. There are various types of Action Buttons that can be setup depending on the business need:

- Plan – System defined action button which can be used as designed. They are primarily used to configure connected excel.
- Download – Also a system defined action button with minimum configurable options. Used to invoke data/report downloads, when pinned to specific widgets
- Widget – Used to invoke rules/actions based on selections within a specific widget
- Rule – This type of action button allows many configuration options. They can be setup and customized as per the business use. A Rule based Action button triggers certain rules or procedures, when executed.

A new action button setup window consists multiple tabs. A few important ones are listed below:

- 1) Config – Used to configure process messages that are displayed based on the triggers of Submit, Cancel, Reset options within an action button form.
- 2) Rule Options – Used to write IBPL rules or Java Scripts, which should be triggered with the execution of an Action Button.

- 3) Field Bindings – These are mechanisms which capture user inputs. Ex: User filter selections can be captured in a field binding within an action button, and rules can be executed only for the filtered data.
- 4) Data Sources – They are used to create a data pool, which uses field binding capture of the user selections and parse them into scope statements, so that resultant dependant member data set is stored. These are further linked to the field binding created for those dependant attributes. Ex: A Product Category can be selected by a user, which can be captured in a data source and result in a list of SKU's against that category.

There are many types of field bindings. A few important ones are explained below:

- 1) Member – Using this field binding, you can directly pull members stored in the Graph-Cube Server, to be displayed within an action button screen and allow user to choose/filter the member data
- 2) Filter – A filter field binding ensures that the report filters are captured and same is displayed within the action button. Options to modify the filters or freeze the same report filters, within the action button, can also be configured.
- 3) Literal – Enables a user to directly input any value. Ex: Literal field bindings can be configured to add new members to the master data.
- 4) Data Source - They capture the user selected filter option for any member and displays the relevant filter data of related members within the report scope.
- 5) Parent Attribute – Enables rule triggers for specific intersections selected by the user within a widget or report

Use Cases:

- 1) Approve the forecast for a specific account (like Walmart) – configure a simple action button with a hard-coded rule and conditions.
- 2) Give the end user to select the Account, SKU and then approve the forecast for these selections – configure an action button with member type field bindings.
- 3) In the Action Buttons window, capture the Report filter selections for SKU & Account and approve the forecast for those. You may or may not allow to change the selections. -- use member field bindings. Field Binding type will be cascadefilter (to cascade the report filters directly to action button filter. Any changes in action button filter will take the screen back to report filter – An action button with a filter field binding will serve this use case.
- 4) Allows end user to enter a new SKU – a literal field binding type can be used for this action button
- 5) In the AB Form, the Account, Prod Category, SKU, Year, Month are to be displayed. List of SKUs must be dependent on the Category selected and the List of Months displayed must be dependent on the Year selected. And the formula should trigger based on the selections. – Action button will use data source.
- 6) End User can select specific intersections from the Report itself and then invoke the button to trigger the formula – Action Button type will be "Widget" and the field binding type is "parentAttribute"

Connected Excel:

Steps to download & install Excel Plug-in

- 1) Create an Action Button with the Type Plan
- 2) Give Name & ToolTip name
- 3) Keep it Global (check-box)
- 4) The Excel Action button can be viewed in Global Header bar
- 5) Click that and select to download the o9 excel add-in
- 6) Close running excel workbooks
- 7) Run the installer
- 8) Open a new workbook
- 9) Click the o9 add-in which will be visible in the workbook & login with tenant user credentials
- 10) In the Reports Model Designer, ensure that the desired Pivots have the radio button 'Excel' selected and saved.
- 11) In the Report designer, create a new Excel Folder, and also a new excel workbook. Add these excel pivots in the newly created workbook.
- 12) While creating a new workbook in the Platform, the check-box 'Is Published', ensures that work done in excel sheet will reflect in the platform.
- 13) Go back to the excel sheet and observe that the pivots list is displayed. Select desired pivot to load the report